

Class 24: Further Applications of the Order of Operations Agreement.  
Simple Interest and Percent Change

Name  
Date  
Arithmetic

### I. Simple Interest

$$A = P + P \times r \times t$$

P = Principal (the money you invest)

r = Percent return written in decimal form.

t = The time your money is earning interest. t is usually in units of years but can also be months or days.

A = The total money you will have (principal and interest) after your money has been invested for a certain period of time.

① You invest \$80 in a savings account at your bank that pays 0.9% simple interest per year. How much money will you have after seven years?

② You take out a \$2,009 loan from a bank that charges seven percent simple interest per day. If you keep the money for 36 days, how much will you owe?

③ You take out a \$600 loan from a bank that charges 2.3% simple interest each month. Assuming that you do not pay any of the money, how much will you owe the bank after 24 months?

④ If you invest \$94 in a bank account that pays 0.5% simple interest yearly, how much money will you have accumulated after six years?

⑤ You borrow \$1500 from a bank that charges eleven percent simple interest yearly. If you don't pay any of the money back for four years, how much will you owe?



⑥ You put \$23.50 in a savings account that pays 0.06% simple interest yearly. How much money will be in this account after 100 years?

### Interest Alone

$$I = P \times r \times t$$

I = Interest

P = Principal

r = Percent return written in decimal form.

t = The time your money is earning interest.

⑦ If you invest \$150 in a bank account that pays you 0.3% simple interest yearly, and you keep the money in the account for three years, how much money will you earn?

⑧ If you borrow \$360 from your brother Bob and he charges you 9% simple interest monthly and you don't pay any of the money back for eight months, how much money in interest will you owe him?

⑨ You loan a business partner \$8,990 at a simple interest rate of 1.2% yearly for one year. How much will he have to pay you in interest?

### II. Percent Change

$$N = F \pm F \times r$$

F = Original number (First number).

r = The percent of increase written as a decimal.

N = The resulting number after the percent change.



⑩ What is 93 increased by 5%?

⑪ What is 7 increased by 36%?

⑫ If 8 is decreased by 97 percent, what is the result?

⑬ What is 23 decreased by 6%?

⑭ What is 6 increased by 90%?

⑮ What is three decreased by 20%?

⑯ If 9.5 is decreased by 2.2%, what does it become?

⑰ If 0.28 is increased by 33 percent, what is the result?

⑱ If 78.2 is increased by 621%, what is the result?



①9 If a store sells a T-shirt for \$12.60, and the sales tax on the shirt is 5%, how much will a customer pay if he buys the shirt?

②0 After using a special fertilizer, the average weight of a farmer's oranges increased by 9.2%. If the average orange on this farmer's land used to be 0.31 pounds, what is the new average weight of an orange?

②1 In a particular country there are 3 million people who own flip phones. After one year the number decreased by 0.19%. How many people own flip phones in the country after the decrease?

Type B Percent Change Problems: Finding the Original Number.

$$F = N \div (1 \pm r)$$

②2 If a number is decreased by 96% and the result is 30, what was the original number?

②3 If a number is increased 6% to 25.44, what was the original number?

②4 If a number is decreased 17% and the result is 0.0249, what was the original number?



②⑤ If a number is increased 21% to 99.22, what was the original number?

②⑥ If a number is decreased 62% to 3.04, what was the original number?

②⑦ If a number is increased 70% and the result is 3.74, what was the original number?

②⑧ If the average length of a lizard is increased by 4%, and the new average length is 54.08 centimeters, what was the original length?

②⑨ If a baby elephant's weight increases 250% from the time it was born and its new weight is 770 lbs, what was his weight at birth?

③⑩ After dropping 7%, the new price of prime steak is \$33.48 per pound. What was the original cost per pound?



Type C Percent Change Problems: Finding the Percent Change

$$r = \frac{\text{Increase}}{(N - F) \div F} \quad \text{or} \quad r = \frac{\text{Decrease}}{(F - N) \div F}$$

31) If 80 is decreased to 30, what is the percent decrease?

32) If 15 is increased to 31, calculate the percent increase.

33) If 40 is increased to 44, what is the percent increase?

34) If 5 is decreased to 3, what is the percent decrease?

35) If 1.5 is decreased to 0.2, calculate the percent decrease.

36) After modifying his recipe, a chef's cookies were reduced in volume from 0.5 ounces to 0.1 ounces. What is the percent decrease of these cookies?

37) Jared increased the area of his posters from  $5.5 \text{ ft}^2$  to  $6 \text{ ft}^2$ . Calculate the percent increase of the area of the posters.